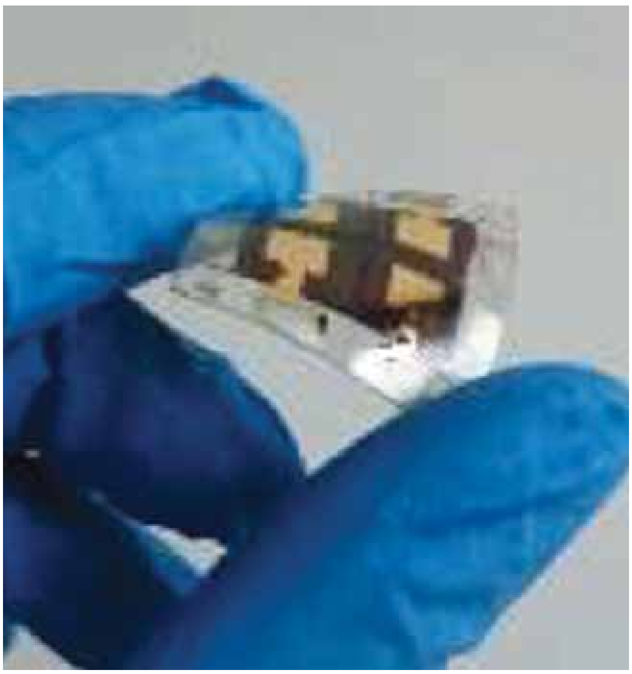
cost-effective batteries from used masks. Thus, medical waste forms the basis for creating batteries. To create a battery of the supercapacitor type, the masks are first disinfected with ultrasound, then dipped in 'ink' made of graphene, which saturates the mask. Then the material is pressed under pressure and heated to 140°C. A separator (also made of mask material) with insulating properties is then placed between the two electrodes made of the new material. It is saturated with a special electrolyte, and then a protective shell is created from the material of medical blister packs (such as paracetamol). The new batteries can be used in household appliances from clocks to lamps.

Perovskite solar cell made with thin paper

A new method that uses a simple sheet of paper to deposit the perovskite films without any expensive equipment at all has been developed by a team from Tor Vergata University and University of Zanjan. The trick to achieve high performance with this remarkably cheap method is to soak the paper applicator in anti-solvent,



Photograph of the solar cell with PET/ITO/SnO2/perovskite/Spiro/Au architecture (Credit: Centre for Hybrid and Organic Solar Energy)

which almost doubles efficiencies compared to when using it dry, reaching 11% on flexible plastic substrates. Paper, compared to other soft applicators, possesses the right porosity and smoothness for deposition of high-quality perovskite films.